



Tecnológico de Monterrey

Proyecto 2 Analizador Sintáctico

Desarrollo de aplicaciones avanzadas de ciencias computacionales

Grupo 501

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Gramática BNF

1. `program` $\rightarrow$ `declaration-list`
2. `declaration-list` $\rightarrow$ `declaration-list declaration` | `declaration`
3. `declaration` $\rightarrow$ `var-declaration` | `fun-declaration`
4. `var-declaration` $\rightarrow$ `type-specifier ID ;` | `type-specifier ID [ NUM ] ;`
5. `type-specifier` $\rightarrow$ `int` | `void`
6. `fun-declaration` $\rightarrow$ `type-specifier ID ( params ) compound-stmt`
7. `params` $\rightarrow$ `param-list` | `void`
8. `param-list` $\rightarrow$ `param-list , param` | `param`
9. `param` $\rightarrow$ `type-specifier ID` | `type specifier ID [ ]`
10. `compound-stmt` $\rightarrow$ `{ local-declarations var-declaration }`
11. `local-declarations` $\rightarrow$ `local-declarations var-declaration` | `empty`
12. `statement-list` $\rightarrow$ `statement-list statement` | `empty`
13. `statement` $\rightarrow$ `expression-stmt` | `compound-stmt` | `selection-stmt` | `iteration-stmt` | `return-stmt`
14. `expression-stmt` $\rightarrow$ `expression ;` | `;`
15. `selection-stmt` $\rightarrow$ `if ( expression ) statement` | `if ( expression ) statement else statement`
16. `iteration-stmt` $\rightarrow$ `while ( expression ) statement`
17. `return-stmt` $\rightarrow$ `return ;` | `return expression ;`
18. `expression` $\rightarrow$ `var = expression` | `simple-expression`
19. `var` $\rightarrow$ `ID` | `ID [ expression ]`
20. `simple-expression` $\rightarrow$ `additive-expression relop additive-expression` | `additive-expression`
21. `relop` $\rightarrow$ `<=` | `<` | `>` | `>=` | `==` | `!=`
22. `additive-expression` $\rightarrow$ `additive-expression addop term` | `term`
23. `addop` $\rightarrow$ `+` | `-`
24. `term` $\rightarrow$ `term mulop factor` | `factor`
25. `mulop` $\rightarrow$ `*` | `/`
26. `factor` $\rightarrow$ `( expression )` | `var` | `call` | `NUM`
27. `call` $\rightarrow$ `ID ( args )`
28. `args` $\rightarrow$ `arg-list` | `empty`
29. `arg-list` $\rightarrow$ `arg-list , expression` | `expression`

Gramática EBNF

1. `program` $\rightarrow$ `declaration-list`
2. `declaration-list` $\rightarrow$ `declaration declaration-list'`
3. `declaration-list'` $\rightarrow$ `declaration declaration-list' | empty`
4. `declaration` $\rightarrow$ `var-declaration | fun-declaration`
5. `var-declaration` $\rightarrow$ `type-specifier ID ; | type-specifier ID [ NUM ] ;`
6. `type-specifier` $\rightarrow$ `int | void`
7. `fun-declaration` $\rightarrow$ `type-specifier ID ( params ) compound-stmt`
8. `params` $\rightarrow$ `param-list | void`
9. `param-list` $\rightarrow$ `param-list , param | param`
10. `param` $\rightarrow$ `type-specifier ID | type specifier ID [ ]`
11. `compound-stmt` $\rightarrow$ `{ local-declarations statement-list }`
12. `local-declarations` $\rightarrow$ `var-declaration local-declarations | empty`
13. `statement-list` $\rightarrow$ `statement statement-list | empty`
14. `statement` $\rightarrow$ `expression-stmt | compound-stmt | selection-stmt | iteration-stmt | return-stmt`
15. `expression-stmt` $\rightarrow$ `expression ; | ;`
16. `selection-stmt` $\rightarrow$ `if ( expression ) statement | if ( expression ) statement else statement`
17. `iteration-stmt` $\rightarrow$ `while ( expression ) statement`
18. `return-stmt` $\rightarrow$ `return ; | return expression ;`
19. `expression` $\rightarrow$ `var = expression | simple-expression`
20. `var` $\rightarrow$ `ID | ID [ expression ]`
21. `simple-expression` $\rightarrow$ `additive-expression relop additive-expression | additive-expression`
22. `relop` $\rightarrow$ `<= | < | > | >= | == | !=`
23. `additive-expression` $\rightarrow$ `term additive-expression'`
24. `additive-expression'` $\rightarrow$ `addop term additive-expression' | empty`
25. `addop` $\rightarrow$ `+ | -`
26. `term` $\rightarrow$ `factor term'`
27. `term'` $\rightarrow$ `mulop factor term' | empty`
28. `mulop` $\rightarrow$ `* | /`
29. `factor` $\rightarrow$ `( expression ) | var | call | NUM`
30. `call` $\rightarrow$ `ID ( args )`
31. `args` $\rightarrow$ `arg-list | empty`
32. `arg-list` $\rightarrow$ `expression arg-list'`
33. `arg-list'` $\rightarrow$ `, expression arg-list' | empty`